

THE TWO LOST NOTES ON PAGE 54 OF RAMANUJAN'S LOST NOTEBOOK

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ABSTRACT. On page 54 of his lost notebook Ramanujan states two families of cubic theta-function identities and, in each family, writes “see note” next to one further claim ($A^3 - B^3$ in the first family, $A^3 + B^3$ in the second). Both notes are lost: Andrews and Berndt record that they have not been preserved. We propose a reconstruction of the two missing identities, obtained by combining the four identities that Ramanujan *did* record on the same page through the elementary factorizations $A^3 \mp B^3 = (A \mp B)^3 \pm 3AB(A \mp B)$, and we verify the resulting closed forms to 2000 coefficients. We then give computational evidence that, unlike the four recorded identities, the two missing combinations admit no representation as a one- or two-term product of theta functions within a large explicitly described search space; this offers an explanation of why these two claims, and only these two, were deferred to a note. All identities in this paper, old and new, are certified by a single elementary integer-arithmetic verification script.

1. INTRODUCTION

As usual, let

$$f(a, b) := \sum_{n=-\infty}^{\infty} a^{n(n+1)/2} b^{n(n-1)/2}, \quad |ab| < 1,$$

and $f(-q) := f(-q, -q^2) = (q; q)_{\infty}$. Page 54 of Ramanujan’s lost notebook [1] contains two groups of cubic identities for theta functions with quotient of arguments q^{15} ; they are treated as Entries 8.2.1 and 8.2.2 in Andrews and Berndt’s second volume [2, pp. 176–178].

Entry 8.2.1. Let $A = f(-q^7, -q^8)$ and $B = qf(-q^2, -q^{13})$. Then

$$(8.2.6) \quad A + B = \frac{f(-q^2, -q^3)}{f(-q, -q^4)} f(-q^5),$$

$$(8.2.7) \quad A - B = f(-q^{2/3}, -q) + q^{2/3} f(-q^3, -q^{12}),$$

$$(8.2.8) \quad A^3 + B^3 = \frac{f(-q^6, -q^9)}{f(-q^3, -q^{12})} f^3(-q^5),$$

$$(8.2.9) \quad (A - B)^3 = \frac{f(-q^2, -q^3) f^3(-q, -q^4)}{f(-q^3, -q^{12})} + q^2 f^3(-q^3, -q^{12}).$$

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After stating these, Ramanujan writes “ $A^3 - B^3$ *see note*.” Andrews and Berndt comment: “Either Ramanujan never wrote this note, or if he did, it has not been preserved” [2, p. 177].

Entry 8.2.2. Let $A = f(-q^4, -q^{11})$ and $B = qf(-q, -q^{14})$. Then

$$(8.2.10) \quad A - B = \frac{f(-q, -q^4)}{f(-q^2, -q^3)} f(-q^5),$$

$$(8.2.11) \quad A + B = -\frac{1}{q^{1/3}} \left\{ f(-q^{1/3}, -q^{4/3}) - f(-q^6, -q^9) \right\},$$

$$(8.2.12) \quad A^3 - B^3 = \frac{f(-q^3, -q^{12})}{f(-q^6, -q^9)} f^3(-q^5),$$

$$(8.2.13) \quad (A + B)^3 = -\frac{1}{q} \left\{ \frac{f(-q, -q^4) f^3(-q^2, -q^3)}{f(-q^6, -q^9)} - f^3(-q^6, -q^9) \right\}.$$

Here Ramanujan writes “ $A^3 + B^3$ *see note*,” and again “he apparently did not write this note, or, if he did, it has been lost” [2, p. 178].

To our knowledge the two missing notes have not been reconstructed in the literature. The cubic identities of page 54 were first proved by Son [4], and Andrews and Berndt state that in this section they “follow precisely Son’s beautiful proofs” [2, p. 173]; both “see note” remarks, with the statements that the notes are lost, appear inside that very section, and the only further content of [4] mentioned there is “some new modular equations” [2, p. 180]. We were unable to find any later source addressing the notes.

This paper does two things. In Section 2 we give what we believe is the natural reconstruction: each missing cube combination follows from the four identities Ramanujan recorded *on the same page*, combined through $A^3 \mp B^3 = (A \mp B)^3 \pm 3AB(A \mp B)$, where the cofactor AB is a single theta-product by the Jacobi triple product. In Section 3 we explain *why* a note was needed at all: within a large explicit search space of theta-monomials, the two missing combinations — in contrast to (8.2.8), (8.2.9), (8.2.12), (8.2.13) — have no one- or two-term product representation. Section 5 describes the verification script that certifies every identity in this paper to 2000 coefficients.

2. RECONSTRUCTION OF THE TWO NOTES

Theorem 1. Let $A = f(-q^7, -q^8)$ and $B = qf(-q^2, -q^{13})$, as in Entry 8.2.1. Then

(1)

$$A^3 - B^3 = \frac{f(-q^2, -q^3) f^3(-q, -q^4)}{f(-q^3, -q^{12})} + q^2 f^3(-q^3, -q^{12}) + 3AB \left\{ f(-q^{2/3}, -q) + q^{2/3} f(-q^3, -q^{12}) \right\},$$

where, by the Jacobi triple product,

$$AB = q f(-q^7, -q^8) f(-q^2, -q^{13}) = q (q^2, q^7, q^8, q^{13}; q^{15})_{\infty} (q^{15}; q^{15})_{\infty}^2.$$

Proof. By the polynomial identity $A^3 - B^3 = (A - B)^3 + 3AB(A - B)$, the claim is precisely (8.2.9) plus $3AB$ times (8.2.7), both of which are proved in

[2, pp. 176–177]. The product form of AB is two applications of the Jacobi triple product $f(a, b) = (-a; ab)_\infty (-b; ab)_\infty (ab; ab)_\infty$. $\square$

Theorem 2. *Let $A = f(-q^4, -q^{11})$ and $B = qf(-q, -q^{14})$, as in Entry 8.2.2. Then*

(2)

$$A^3 + B^3 = -\frac{1}{q} \left\{ \frac{f(-q, -q^4) f^3(-q^2, -q^3)}{f(-q^6, -q^9)} - f^3(-q^6, -q^9) \right\} + \frac{3AB}{q^{1/3}} \left\{ f(-q^{1/3}, -q^{4/3}) - f(-q^6, -q^9) \right\},$$

where

$$AB = q f(-q^4, -q^{11}) f(-q, -q^{14}) = q (q, q^4, q^{11}, q^{14}; q^{15})_\infty (q^{15}; q^{15})_\infty^2.$$

Proof. $A^3 + B^3 = (A + B)^3 - 3AB(A + B)$; substitute (8.2.13) and (8.2.11), proved in [2, p. 178]. $\square$

Both right-hand sides were verified coefficient-by-coefficient to 2000 terms (Section 5). We emphasize what is, and what is not, claimed: the two theorems are elementary consequences of Ramanujan's own four recorded identities, and in that sense they are certainly results Ramanujan possessed. Our claim is that (1) and (2) are the *content* of the two lost notes in the only sense now recoverable: they express $A^3 - B^3$ and $A^3 + B^3$ in closed form using exactly the functions appearing on page 54, and (Section 3) no representation of the same one- or two-term shape as (8.2.8)–(8.2.9) or (8.2.12)–(8.2.13) exists to be found.

3. WHY A NOTE WAS NECESSARY: A NONEXISTENCE COMPUTATION

Each of the four cube identities Ramanujan recorded on page 54 is a one- or two-term combination of monomials in theta functions. It is natural to ask whether $A^3 - B^3$ (Entry 8.2.1) and $A^3 + B^3$ (Entry 8.2.2) also possess such a form, which Ramanujan would simply have recorded in the note. The evidence is that they do not.

3.1. Non-periodicity of the quadratic cofactors. Write $A^3 + B^3 = (A + B)(A^2 - AB + B^2)$ and $A^3 - B^3 = (A - B)(A^2 + AB + B^2)$. A power series with integer coefficients and constant term 1 has a unique representation $\prod_{n \geq 1} (1 - q^n)^{-c_n}$. If (c_n) is periodic with period M , the series equals $\prod_{a=1}^M (q^a; q^M)_\infty^{-c_a}$, a generalized eta-quotient, i.e. a finite product of theta functions of the kind occurring throughout page 54 (Jacobi triple product); conversely every such product has periodic (c_n) . Computing (c_n) for $n \leq 380$ by the logarithmic-derivative recursion we find, in each entry, a sharp split:

Entry	combination	exponent sequence (c_n)
8.2.1	$A^2 - AB + B^2$ (factor of the recorded (8.2.8))	periodic mod 15
8.2.1	$A^2 + AB + B^2$ (factor of the lost $A^3 - B^3$)	not periodic
8.2.2	$A^2 + AB + B^2$ (factor of the recorded (8.2.12))	periodic mod 15
8.2.2	$A^2 - AB + B^2$ (factor of the lost $A^3 + B^3$)	not periodic

(“Not periodic” means: no period ≤ 60 holds for $n \leq 380$; the periodic cases were checked from $n = 1$.) Thus in each family the recorded cube identity is the one whose quadratic cofactor is a generalized eta-quotient, and the deferred one is the one whose cofactor is not. This already rules out any identity of the shape $A^3 \mp B^3 = (\text{linear factor known on p. 54}) \times (\text{generalized eta-quotient})$.

3.2. Systematic search over theta-monomial bases. We searched for representations

$$A^3 - B^3 = \sum_{i=1}^r c_i q^{s_i} M_i, \quad r \leq 2, \quad c_i \in \{\pm 1, \pm 2, \pm 3\}, \quad s_i \in \{0, \dots, 3\},$$

where each M_i is a monomial of total degree 3 with exponents in $\{-1, 1, 2, 3\}$ in at most three functions drawn from the 12-function basis

$$f(-q, -q^4), f(-q^2, -q^3), f(-q^3, -q^{12}), f(-q^6, -q^9), f(-q^5), \\ f(-q, -q^{14}), f(-q^2, -q^{13}), f(-q^4, -q^{11}), f(-q^7, -q^8), f(-q), f(-q^3), f(-q^{15})$$

— every theta function occurring in Entries 8.2.1–8.2.2 together with the relevant η -type functions (2344 monomials; hash-matching on the first 80 coefficients, surviving candidates confirmed on 260). The search was repeated in the $q^{1/3}$ -lattice (setting $q \mapsto q^3$ and adjoining the functions $f(-q^{a/3}, -q^{b/3})$ occurring in (8.2.7) and (8.2.11): 20 basis functions, 11800 monomials, shifts $s_i \in \{0, \dots, 9\}$), and with the three-term shape $c_1 q^{s_1} M_1 + c_2 q^{s_2} M_2 \pm 3q^s AB M_3$ suggested by Theorems 1–2.

Control of the instrument. Run on the recorded combinations, the same search re-finds Ramanujan’s identities exactly: for $(A - B)^3$ of Entry 8.2.1 it returns the two-term representation (8.2.9), and in the $q^{1/3}$ -lattice it returns (8.2.7) for $A - B$.

Result. For $A^3 - B^3$ (Entry 8.2.1) and $A^3 + B^3$ (Entry 8.2.2) the search returns nothing beyond trivial rewritings of the definitions. Within the declared space — which contains every identity actually appearing on page 54 — the two lost combinations have no one- or two-term product form.

We state the scope honestly: this is a finite search over an explicit basis and coefficient range, not a proof of nonexistence in general; a completeness argument via the theory of modular functions on $\Gamma_0(15)$ (the natural next step) is left open.

4. THE MIRROR SYMMETRY, AND A REMARK ON THE NOTE

The two entries are mirror images: under the exchange $f(-q^3, -q^{12}) \leftrightarrow f(-q^6, -q^9)$, $f(-q, -q^4) \leftrightarrow f(-q^2, -q^3)$, identity (8.2.8) corresponds to (8.2.12) and (8.2.9) to (8.2.13). In each family Ramanujan recorded the cube combination with the generalized eta-quotient cofactor and deferred the other to a note. The parallel structure of the two deferrals — the *same*

obstruction on both sides of the mirror — suggests the notes were to contain the same kind of statement for both entries; the factorizations of Theorems 1–2 are the canonical such statement, since $(A \mp B)$ and $(A \mp B)^3$ are exactly the quantities (8.2.7)/(8.2.9) and (8.2.11)/(8.2.13) that Ramanujan placed immediately above each remark. Whether Ramanujan intended precisely this, or a representation of a shape we have not searched, cannot of course be decided; we label this paragraph as interpretation, not theorem.

5. THE VERIFICATION SCRIPT

All ten identities of this paper — (8.2.6)–(8.2.13) and (1)–(2) — are certified by one self-contained Python script `verifier.py` (about 130 lines, no dependencies). It works in the variable $y = q^{1/3}$ with exact integer arithmetic; every identity involving a quotient is verified after cross-multiplication, so no division of power series occurs. Each identity is checked to 400 coefficients of q by default, and the run reported here used 2000 coefficients (6003 powers of y ; total runtime under one second). The script prints one MOS/FARQ (match/mismatch) line per identity and exits nonzero on any mismatch; the archived run prints 10/10 MOS.

DATA AVAILABILITY

The verification script and the search scripts of Section 3 are archived at Zenodo: <https://doi.org/10.5281/zenodo.23030497>.

REFERENCES

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- [3] B. C. Berndt, *Ramanujan's Notebooks, Part III*, Springer, New York, 1991 (Chapter 20, Entries 10(i)–(iv)).
- [4] S. H. Son, *Cubic identities of theta functions*, *Ramanujan J.* **2** (1998), 303–316, doi:10.1023/A:1009751614537.